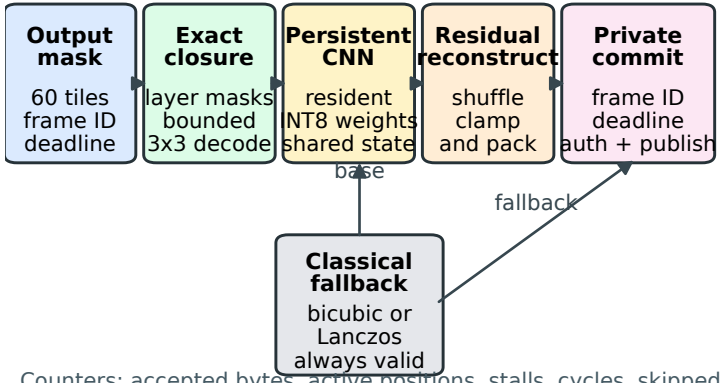
Exact mask-driven sparse refinement with commit-safe fallback



Counters: accepted bytes, active positions, stalls, cycles, skipped work, and terminal outcome